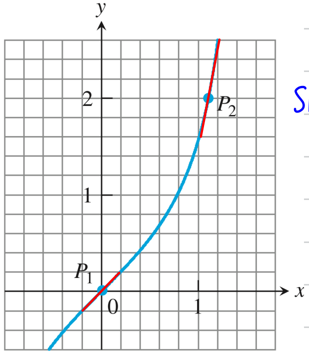
Section 3.1 Atthaseth Kanokwutthithumrong 125010006 T03

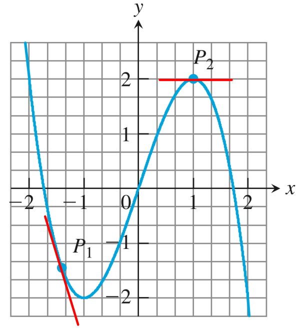
1.



Slope at $P_2 = 5$

Slope at $P_1 = 1$

2.



Slope at $P_2 = 0$

Slope at $P_1 = -3$

23. a) Instantaneous rate of change of number of yeast cells at time $t=5$ hours

Unit : cells / hour

b) $P'(3)$ is larger because the slope of graph at $t=3$ is steeper than the slope at $t=2$

c) $$P'(t) = \lim_{h\to 0} \frac{P(t+h)-P(t)}{h} = \lim_{h\to 0} \frac{(6.10(t+h)^2 - 9.28(t+h)+16.43)-(6.10t^2-9.28t+16.43)}{h}$$

$$= \lim_{h\to 0} \frac{6.10(2th+h^2)-9.28h}{h} = \lim_{h\to 0} 6.10(2t+h)-9.28 = 12.2t-9.28$$

$\therefore P'(5) = 12.2 \cdot 5 - 9.8 = 51.2$ cells/hour

33. Slope of tangent line of graph $y = mx+b$ at point (x_0, mx_0+b) is

$$\lim_{h\to 0} \frac{(m(x_0+h)+b)-(mx+b)}{h} = \lim_{h\to 0} \frac{mh}{h} = m$$

The equation of tangent line at (x_0, mx_0+b) is

$$y-(mx_0+b) = m(x-x_0)$$
$$y - mx_0 - b = mx - mx_0$$
$$y = mx+b$$

Hence, the line $y=mx+b$ is its own tangent line at any point (x_0, mx_0+b)

41. $\frac{dy}{dx} = \frac{1}{5}x^{-4/5}$ $\therefore$ at $x=0$, $\frac{dy}{dx} = \infty$ or it has a vertical tangent at $x=0$.

47. y is continuous at $x=0$, and has a vertical tangent at $x=0$.

$x<0$: $$\lim_{h\to 0} \frac{-\sqrt{|x+h|}-(-\sqrt{|x|})}{h} = \lim_{h\to 0} \frac{\sqrt{|x|}-\sqrt{|x+h|}}{h} = \lim_{h\to 0} \frac{\sqrt{-x}-\sqrt{-x-h}}{h} = \lim_{h\to 0} \frac{1}{\sqrt{-x}+\sqrt{-x-h}} = \frac{1}{2\sqrt{-x}}$$

$\therefore \lim_{x\to 0^-} y' = \lim_{x\to 0} \frac{1}{2\sqrt{-x}} = \infty$

$x>0$: $$\lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} = \lim_{h\to 0} \frac{1}{\sqrt{x+h}+\sqrt{x}} = \frac{1}{2\sqrt{x}}$$

$\therefore \lim_{x\to 0^+} y' = \lim_{x\to 0^+} \frac{1}{\sqrt{2x}} = \infty$

$\therefore \lim_{x\to 0} y' = \infty$ so at $x=0$, y has a vertical tangents